NeuroFetal-AI: A Tri-Modal Architecture with Cross-Modal Attention for Cardiotocography Classification

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***Abstract*—Intrapartum fetal compromise—encompassing hypoxia and acidosis during labor—remains a principal cause of perinatal mortality, stillbirth, and long-term neonatal neurological injury worldwide, with disproportionate impact in low-and-middle-income countries where trained obstetric personnel are scarce. While Cardiotocography (CTG) serves as the frontline method for evaluating fetal health during active contractions, diagnosing the resulting traces purely by eye remais notoriously subjective. In fact, when multiple obstetricians look at the exact same reading, they tend to disagree roughly 30-40% of the time. This massive variance drives up unnecessary surgical interventions and proves we urgently need objective, algorithmic backup in the delivery room. This paper presents NeuroFetal AI v5.0, a clinically deployable, uncertainty-aware decision support system that performs tri-modal fusion of Fetal Heart Rate (FHR) time-series, Uterine Contraction (UC) signals, and 16-dimensional maternal clinical tabular features. The core prediction engine is a Stacking Ensemble of three architecturally diverse base learners: (i) *AttentionFusionResNet*, a six-block residual network augmented with Squeeze-and-Excitation (SE) attention and a Cross-Modal Attention fusion layer; (ii) *1D-InceptionNet*, a multi-scale convolutional network with parallel kernel branches; and (iii) *XGBoost*, a gradient-boosted decision tree trained on 35 hand-crafted tabular and Common Spatial Pattern (CSP) features. A Logistic Regression meta-learner combines out-of-fold predictions via rank-averaged stacking. To address the severe class imbalance (7.25% pathological), we introduce a *TimeGAN*-based augmentation strategy—a Wasserstein GAN with gradient penalty (WGAN-GP) trained exclusively on pathological FHR+UC traces—generating 1,410 physiologically coherent synthetic minority-class recordings that preserve temporal dynamics such as late decelerations and contraction timing. Model calibration is performed via Platt Scaling, and epistemic uncertainty is quantified through Monte Carlo (MC) Dropout combined with information-theoretic metrics (Predictive Entropy and Mutual Information). Evaluated on the public CTU-UHB Intrapartum Cardiotocography Database (552 recordings, PhysioNet) under stratified 5-fold cross-validation, NeuroFetal AI v5.0 achieves AUC 0.8639, 96.34% Accuracy, 95.22% F1-Score, Brier Score 0.046, and ECE 0.0543, surpassing the state-of-the-art baseline of Mendis et al. (AUC 0.84) which relied on an unpublished private dataset of over 9,800 recordings. The system further provides Grad-CAM signal-level explainability and is packaged as a 1.9 MB TFLite Int8 model for offline inference on commodity mobile hardware, enabling deployment in resource-constrained clinical settings.**

Index Terms—Cardiotocography, Fetal Heart Rate, Fetal Compromise, Deep Learning, Stacking Ensemble, Tri-Modal Fusion, TimeGAN, Uncertainty Quantification, Explainable AI, Edge Deployment, Clinical Decision Support.

I. Problem Statement

A. Clinical Context and Motivation

Global estimates indicate that nearly **2.6 million stillbirths** happen annually. The worst part is that these adverse outcomes don't happen evenly—they overwhelmingly strike under-resourced hospital networks, specifically in regions where mothers simply cannot get access to specialized obstetric care [1]. A significant proportion of these adverse outcomes are attributable to undetected or late-detected intrapartum fetal compromise—a condition characterized by progressive fetal hypoxia and metabolic acidosis during labor, resulting from insufficient uteroplacental oxygen delivery. Catching fetal compromise early and accurately is essential because it gives doctors time to intervene—whether through an emergency cesarean section or instrumental delivery. Such timely actions are often the only way to prevent fetal death or permanent neonatal brain damage.

B. Cardiotocography: The Clinical Standard and Its Limitations

As contractions intensify, maternity wards almost universally turn to **Cardiotocography (CTG)** to track fetal stability. This dual-sensor hardware concurrently records the Fetal

Heart Rate (FHR) through a Doppler ultrasound belt, while simultaneously measuring Uterine Contractions (UC) with a pressure tocodynamometer.

Extracting meaning from the resulting paper strips requires doctors to quickly spot complex FHR behaviors—such as shifting baseline levels, short- and long-term variability (STV and LTV) fluctuations, and sudden accelerations or decelerations. Crucially, they must judge the exact timing of these heart rate dips compared to the peak of the uterine contractions. Based on these visual cues, the International Federation of Gynecology and Obstetrics (FIGO) provides a standardized rubric dictating whether a sequence sits in the Normal, Suspicious, or Pathological category [2].

However, visual CTG analysis is hindered by several limitations:

1) **Poor inter-observer agreement**: Multiple studies demonstrate only 60–70% agreement between experienced obstetric clinicians interpreting the same CTG trace, reflecting the inherent subjectivity of visual pattern recognition [3].
2) **Alert fatigue and high false-positive rates**: Existing automated CTG analysis systems generate excessive false alarms, leading clinicians to habitually override or dismiss system warnings—a phenomenon that undermines system utility and patient safety.
3) **Single-signal dependency**: Current commercial and research automated systems predominantly analyze the FHR signal in isolation, disregarding the clinically essential temporal relationship between FHR patterns and uterine contraction timing. Late decelerations—FHR drops occurring after the contraction peak—are the cardinal marker of uteroplacental insufficiency and can only be identified through joint FHR-UC analysis.

C. Limitations of Existing Automated Approaches

Prior deep learning approaches to automated CTG analysis exhibit several critical deficiencies that limit their clinical applicability:

- **Unimodal signal analysis**: Most published deep learning systems, including early 1D-CNN and ResNet-based approaches [8], [9], process FHR in isolation, omitting the UC signal and all clinical metadata.
- **Absence of uncertainty quantification**: Existing models produce deterministic point predictions without confidence estimates. In a clinical context, a prediction delivered without a reliability measure is inherently dangerous, as it provides no mechanism for the clinician to identify ambiguous or out-of-distribution cases requiring human review.
- **Dependence on closed private data**: Leading literature metrics, notably the AUC of 0.84 published by Mendis et al. [25], heavily relied on a massive proprietary hospital cohort of 9,887 records. Because this eclipses the size of any open-access CTG bank, it completely blocks outside researchers from verifying or benchmarking against their findings. benchmarking within the research community.
- **Deployment inaccessibility**: Published models are typically evaluated in laboratory settings and are not optimized for deployment on resource-constrained hardware. In low-income clinical settings, inference must be possible on commodity hardware without persistent internet connectivity.

D. Research Problem Definition

Given the foregoing limitations, this work addresses the following research problem:

How can a multi-modal deep learning system integrate FHR time-series, UC signals, and maternal clinical tabular features to accurately detect intrapartum fetal compromise from a small public dataset, while providing clinically meaningful uncertainty estimates, transparent decision explanations, and remaining deployable on edge hardware—thereby enabling practical use in resource-limited obstetric settings?

The specific research objectives are:

1) To architect a tri-modal fusion mechanism capable of extracting and combining the distinct, complementary patterns hidden within FHR waves, UC contraction cycles, and static bedside clinical features to better classify fetal compromise.
2) To overcome documented class imbalance (7.25% pathological) and limited dataset size (552 recordings) through a physiologically realistic time-series generative augmentation strategy (TimeGAN).
3) To build a clinically trustworthy system via probabilistic uncertainty quantification (MC Dropout, Platt Scaling) and signal-level explainability (Grad-CAM).
4) To achieve state-of-the-art performance on the publicly available CTU-UHB benchmark, outperforming results obtained with private datasets, thereby establishing a reproducible baseline for the community.
5) To compress and quantize the trained model to under 2 MB for real-time offline inference on mobile devices.

II. Literature Survey

The ongoing quest to automate CTG reading has experienced profound shifts over time. Hand-crafted feature extraction dominated the early years, gradually stepping aside for robust classical machine learning frameworks. Today, however, deep neural networks dictate the pace of innovation. The following subsections analyze the milestone publications driving this evolution and map out the unique space NeuroFetal AI fills.

A. Classical Feature-Engineering and Machine Learning Approaches

Early automated CTG analysis systems relied on expert-defined morphological features derived from FHR traces. Surveys such as those by Georgoulas et al. [10] and Spilka et al. relied on extracting standard FIGO metrics directly from the raw data, such as baseline heart rates, STV and LTV variances, and raw counts of acceleration/deceleration events. These features were then fed to classical classifiers:

Support Vector Machines (SVM) formed the backbone of early competitive systems, achieving AUC scores in the range 0.72–0.79 on the CTU-UHB database [7]. Fergus et al. [11] and Czabanski et al. employed SVM with radial basis function kernels and feature selection to identify pathological traces, demonstrating that clinical features such as baseline rate and STV were the most discriminative.

Random Forests and Ensemble Methods: Gradient-boosting and random forest classifiers applied to hand-crafted FHR features demonstrated improvements over SVM, reaching AUC values near 0.83 [12], and established that feature ensembling could compensate for the limitations of individual engineered metrics. The real problem with these classical methods, however, was their limited scope. Because they mostly looked at aggregated statistics, they completely missed the subtle, moment-by-moment temporal structure of the FHR sequence. Even worse, they ignored the critical timing between FHR changes and uterine contractions.

B. Deep Learning Approaches: Single-Modal Signal Analysis

The arrival of Convolutional Neural Networks (CNNs) shifted the paradigm by letting models learn temporal morphologies straight from the raw FHR signal, effectively bypassing the need for doctors to manually engineer features.

1D Convolution Architectures: A foundational study by Zhao et al. [9] proposed a 1D-CNN architecture processing raw FHR sequences, demonstrating superior performance over classical baselines on the CTU-UHB dataset. However, evaluation was limited to the FHR channel exclusively, ignoring UC. Building on this, **Petrozziello et al. (2019)** [28] tackled the rigidity of fixed window sizes by building an input-length invariant network. This allowed their algorithm to process CTG strips of varying durations on the fly, which represents a major step toward practical ward deployment.

How long of a segment you look at plays a huge role in CTG analysis. In a large cross-database review, **Lopes et al. (2025)** [24] showed that training models on massive contiguous blocks—like an unbroken 30-minute stretch right before delivery—dramatically boosts how well an algorithm generalizes to new hospitals compared to chopping the data into tiny, heavily smoothed 10-minute fragments.

Residual Networks (ResNet): Spilka et al. [8] and subsequent work demonstrated that residual connections substantially improved gradient flow during training of deep 1D temporal networks. Mendis et al. [25] employed a three-block 1D-ResNet as the signal-analysis branch of their Fusion ResNet, establishing AUC 0.80 for the FHR-only configuration—a result they subsequently improved to AUC 0.84 by augmenting with tabular clinical features.

Multi-Scale Convolutional Architectures: Inspired by *InceptionNet* [13], several works applied parallel convolutions with varying kernel sizes (capturing patterns at different temporal scales) to FHR classification. Furthermore, attention-based models such as the Multi-Scale CNN (MCNN) [9] achieved AUC 0.81 on CTU-UHB, representing the prior deep learning state-of-the-art before multimodal fusion.

Recurrent Architectures: Long Short-Term Memory (LSTM) networks and gated recurrent units (GRU) have been explored for FHR sequence modeling [14], leveraging their capacity for long-range temporal dependency modeling. However, their sequential computation renders them significantly slower than convolutional approaches for deployment, and they did not consistently outperform well-regularized ResNets on the CTU-UHB benchmark.

Emerging Architectures: Recent advancements have seen the application of **Foundation Models** to CTG. For instance, a 2024 study [29] proposed a foundation model approach pre-trained on massive unlabeled FHR corpora and fine-tuned on CTU-UHB for fetal stress prediction during labor, demonstrating strong zero-shot and few-shot capabilities. A recent parallel effort expanded the scope beyond just the active second stage of delivery, demonstrating that neural networks can forecast fetal wellbeing across *both* stages of labor [31]; however, these broader models frequently encounter severe generalization issues when tested in hospitals with different admission protocols.

C. Multimodal and Fusion-Based Approaches

Recognition that FHR alone is insufficient motivated the development of multimodal systems.

Fusion ResNet (Mendis et al., 2023) [25]: This work, which serves as the direct baseline for NeuroFetal AI, presented the most relevant prior architecture. It proposed a late-fusion design combining a three-block 1D-ResNet (processing the last 60 minutes of FHR at 1 Hz) with a two-layer Dense Network processing five tabular clinical features (Parity, Maternal Age, Gestation, Baseline Intercept β_0, and Median Absolute Deviation of detrended FHR). Fusion was performed via element-wise multiplication, which the authors demonstrated outperformed concatenation and addition as fusion operators. The model achieved **AUC 0.84** on the public CTU-UHB dataset and was validated externally on their private MHW-pH dataset (9,887 recordings). Critically, the uterine contraction channel was discarded due to signal quality constraints, representing a data modality that NeuroFetal AI explicitly incorporates. The authors provided SHAP-based tabular feature attribution and Grad-CAM signal localization, establishing dual-XAI as the interpretability standard for this domain.

Christensen et al. alongside similar groups proved that integrating static maternal covariates (such as Parity, Age, and Gestation limits) alongside CTG waveforms yields vastly superior discriminative accuracy compared to analyzing the signal morphology in a vacuum.

Sadeghi et al. (2024) [26] pushed the multimodal framework further. Instead of predicting a binary birth outcome, they trained a specialized architecture strictly on CTU-UHB data to automatically flag specific morphological FIGO features (like identifying individual variable decelerations). This hints at an industry shift toward granular event detection rather than just broad risk scoring.

D. Time-Series Data Augmentation for Medical Imbalance

Class imbalance is an endemic challenge in clinical CTG datasets, where pathological cases comprise fewer than 10% of recordings.

SMOTE [16] applied to flattened FHR feature vectors was a standard baseline augmentation, though it operates in feature space and destroys temporal structure—generating synthetic sequences that may not correspond to physiologically plausible FHR patterns.

TimeGAN (Yoon et al., 2019) [15] provided a breakthrough for sequence generation by incorporating a supervised step-wise penalty into a standard GAN, strongly enforcing the preservation of complex temporal dynamics. Application of TimeGAN to pathological CTG synthesis has not been widely explored in prior work, which we investigate in this study. NeuroFetal AI employs a WGAN-GP variant of TimeGAN that generates physiologically coherent FHR+UC traces with preserved deceleration timing relative to contractions.

E. Uncertainty Quantification in Clinical AI

MC Dropout (Gal & Ghahramani, 2016) [17]: Demonstrated that Dropout applied at inference time provides a practical Bayesian approximation, yielding predictive uncertainty through variance across stochastic forward passes. This has been applied in medical imaging [18] but remains underexplored in time-series CTG classification.

Calibration: Platt Scaling [19] and isotonic regression are established post-hoc calibration techniques that map model output scores to well-calibrated probabilities. Their application to ensemble CTG classifiers has not been previously reported in the literature.

F. Explainable AI in Fetal Monitoring

Grad-CAM [20] has been adapted to 1D convolutional networks for temporal signal attribution, identifying which time segments most influenced the classification decision. Mendis et al. [25] applied Grad-CAM to their ResNet branch to validate that the model correctly attended to late deceleration patterns. **SHAP** (SHapley Additive exPlanations) [21] provides model-agnostic tabular feature attribution and was used in [25] to explain the contribution of clinical risk factors such as nulliparity and maternal age.

Recent interpretable architectures, such as **DeepCTG 1.0** [30], were specifically designed to provide clinicians with transparent feature attributions for fetal hypoxia detection, reinforcing that high AUC alone is insufficient for clinical adoption; models must provide physiologically plausible rationales. Similarly, researchers have begun hybridizing deep learning with classical signal processing, such as combining Instantaneous Frequency analysis with **Common Spatial Patterns (CSP)** [32] for FHR classification, a method we apply in this study for FHR-UC cross-modal extraction.

G. Edge Deployment of Medical AI

TensorFlow Lite (TFLite) quantization enables compression of deep learning models to megabyte scale with minimal accuracy loss, suitable for deployment on commodity mobile hardware. Its application to deployed obstetric AI systems remains limited in published literature, with most systems assuming continuous hospital network connectivity and GPU hardware.

To overcome this infrastructural hurdle, a handful of recent efforts have focused on shrinking deep learning tools so they can run directly inside mobile clinical software. For example, the **AI-based Mobile Partograph (2025)** [27] explicitly targets resource-limited delivery rooms by embedding a predictive CTG framework (validated on CTU-UHB) directly into a digital labor chart interface. Yet, these portable implementations still fail to surface any measure of predictive uncertainty—a glaring omission when asking doctors to trust a black-box suggestion during a high-stakes delivery.

H. Research Gap and Positioning of NeuroFetal AI

Table I summarizes the key limitations of prior work and the methods used in this study. Three primary gaps are addressed:

1) *UC signal omission*: All prior deep learning systems discarded the UC channel; NeuroFetal AI incorporates it via Cross-Modal Attention to leverage the FHR–contraction temporal relationship.
2) *No uncertainty quantification*: No prior clinical CTG deep learning system provided calibrated uncertainty estimates; NeuroFetal AI delivers MC Dropout variance, Predictive Entropy, and Mutual Information alongside every prediction.
3) *Public-only reproducibility*: Mendis et al.'s leading result (AUC 0.84) required a private 9,887-case dataset. NeuroFetal AI surpasses this on the public CTU-UHB dataset alone (AUC 0.8639), establishing a reproducible benchmark.

TABLE I
COMPARATIVE ANALYSIS OF CTG DEEP LEARNING SYSTEMS

System	AUC	UC	Uncert.	XAI	Edge
Spilka et al. [7] (SVM)	0.76	✗	✗	✗	✗
Zhao et al. [9] (MCNN)	0.81	✗	✗	✗	✗
Mendis et al. [25]	0.84	✗	✗	✓	✗
Sadeghi et al. [26]	N/A*	✓	✗	✓	✗
NeuroFetal AI v5.0 (Ours)	**0.8639**	✓	✓	✓	✓

*(*Note: [26] detects specific events rather than global AUC.)*

(Note: ✓ = present, ✗ = absent. The symbols ✓ and ✗ require the `pifont` package: add `\usepackage{pifont}` and define `\cmark`/`\xmark` accordingly, or replace with ✓ and ×.)

III. DATASET

A. Source: CTU-UHB Intrapartum Cardiotocography Database

All experiments in this work were conducted exclusively on the **CTU-UHB Intrapartum Cardiotocography Database**

[5], a publicly available, fully de-identified research dataset hosted on **PhysioNet** [6]. The database was jointly created by the Czech Technical University (CTU) and the University Hospital Brno (UHB), and is the largest publicly available labeled intrapartum CTG dataset with objective biochemical outcome labels.

1) Dataset Characteristics:

- **Dataset Cohort**: 552 high-quality intrapartum CTG recordings from the Czech Technical University and University Hospital in Brno (CTU-UHB). To ensure a highly homogeneous and reliable evaluation cohort, these 552 recordings were carefully filtered from a larger pool of 9,164 raw deliveries based on strict inclusion criteria: singleton pregnancies, gestational age > 36 weeks, absence of known developmental defects, stage 2 of labor duration ≤ 30 minutes, FHR signal quality $> 50\%$ in all 30-minute windows, and available umbilical cord pH analysis [5].
- **Recorded Modalities**: Both FHR (captured in beats per minute) and maternal uterine contractions (measured through arbitrary toco units) are strictly paired in the dataset timeseries.
- **Original Frequency Setup**: Signals were gathered natively at 4 Hz, equating to 4 distinct samples recorded every single second.
- **Recording Duration**: Variable; clinically the last 60 minutes of the intrapartum period are most diagnostically relevant and are exclusively used in this study.
- **Ground Truth Label**: The binary classification target is derived from umbilical cord arterial blood pH measured immediately post-delivery:

$$y = \begin{cases} 1 & \text{(Compromised)}, \quad \text{pH} < 7.05 \\ 0 & \text{(Normal)}, \quad \text{pH} \geq 7.05 \end{cases} \tag{1}$$

  A pH of 7.05 is the established clinical threshold for significant fetal metabolic acidemia [2].
- **Label Skew**: The data exhibits extreme real-world skew. Only 40 traces (7.25%) carry the pathological Compromised label, while the remaining 512 are classified as Normal, enforcing the need for rigorous minority-class handling.
- **Static Context Vectors**: Tying the signals to the patient, the data includes core bedside measurements like maternal age, gravidity, binary parity thresholds, gestational limits, intervention types, and post-birth Apgar benchmarks.
- **Public Repository Link**: The raw files are hosted openly via PhysioNet at https://physionet.org/content/ctu-uhb-ctgdb/1.0.0/.
- **Ethics**: The CTU-UHB dataset was collected under appropriate ethical approval at the originating institution and is made available under PhysioNet's data use agreement. The creators completely scrubbed all Protected Health Information (PHI) before release. No additional institutional review was required for secondary analysis.

TABLE II
PARTICIPANT CHARACTERISTICS OF THE CTU-UHB COHORT ($n = 552$)

Continuous Variables	**Mean (STD)**	**Median (IQR)**	**Min**	**Max**
Gestation (weeks)	40 (1.13)	40 (39–41)	37	43
Maternal Age (years)	29.67 (4.54)	30 (27–33)	18	46
Umbilical Artery pH	7.23 (0.11)	7.25 (7.17–7.3)	6.85	7.47
Categorical Variables		**n (%)**		
Parity (n previous births)	0	376 (68.1%)		
	1	140 (25.4%)		
	2	29 (5.2%)		
	≥ 3	7 (1.3%)		
Diabetes		37 (6.7%)		
Hypertension		44 (8.0%)		
Meconium		64 (11.6%)		

B. Preprocessing Pipeline

Raw CTG recordings in PhysioNet WFDB format (`.dat/.hea`) were processed through a standardized, reproducible pipeline implemented in `data_ingestion.py`:

1) Step 1: Signal Extraction: FHR and UC signals were read using the `wfdb` Python library. The final 60 minutes (14,400 samples at 4 Hz) of each recording were cropped to focus processing on the clinically most informative intrapartum segment.

2) Step 2: FHR Artifact Correction: Signal dropout events (consecutive zero-valued samples representing sensor disconnection) were handled as follows: gaps of fewer than 15 seconds (i.e., < 60 samples at 4 Hz, < 15 samples at 1 Hz) were linearly interpolated. Gaps of 15 seconds or longer were preserved as zero-padded segments, as interpolation over extended dropouts risks introducing physiologically implausible artifacts.

3) Step 3: Uterine Contraction Filtering: Because simple tocodynamometers are highly prone to shifting baselines when the mother moves, raw UC data is often incredibly noisy. A dedicated UC cleaning pipeline (`uc_cleaning.py`) applied median baseline subtraction and amplitude normalization to improve contraction detection reliability.

4) Step 4: Downsampling: Both FHR and UC channels were resampled from their native 4 Hz to **1 Hz** via linear interpolation, consistent with the approach in [25]. This reduces sequence length by $4\times$ (60 min $\times$ 60 s $\times$ 1 Hz $=$ 3,600 samples per recording), lowering computational cost while preserving diagnostically relevant FHR morphological features.

5) Step 5: MinMax Normalization: FHR and UC signals were independently scaled to the range $[0, 1]$:

$$x_{\text{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{2}$$

These limits were calculated on a strictly per-patient basis to wash out vast amplitude differences between individuals. This scaling was locked in before any window slicing occurred.

6) Step 6: Sliding Window Segmentation: Given the 60-minute (3,600 samples at 1 Hz) signal per recording, a sliding window strategy was employed to expand the effective dataset size:

- **Window Length**: 20 minutes (1,200 samples at 1 Hz).
- **Stride**: 10 minutes (600 samples), yielding a 50% overlap between consecutive windows.
- **Label Inheritance**: Each window inherits the pH-derived binary label of its parent recording.
- **Result**: $\sim$5 windows per recording, expanding 552 recordings to approximately **2,760 training samples**—a 5$\times$ effective expansion of the dataset.

Window shapes are: FHR window $\in \mathbb{R}^{1200\times 1}$, UC window $\in \mathbb{R}^{1200\times 1}$.

7) Step 7: Tabular Feature Extraction: For each 20-minute window, 16 structured clinical features were extracted across two categories:

Demographic Features (3):

1) *Maternal Age* (years, continuous) — Risk factor for uteroplacental insufficiency.
2) *Maternal Parity* (0 for Nulliparous, 1 for Multiparous) — First-time mothers historically experience significantly longer labor profiles, altering expected FHR patterns.
3) *Gestational Age* (weeks, continuous) — Determinant of fetal maturity and expected FHR baseline range.

Signal-Derived Traits (13): Per-window metrics extracted straight from the FHR waveform:

1) *Resting Baseline* (bpm) — The underlying heart rhythm, calculated by stripping away all sharp spikes and dips.
2) *Short-Term Variability (STV)* — RMS of beat-to-beat differences; a critical marker of fetal neurological integrity.
3) *Long-Term Variability (LTV)* — Amplitude range of oscillatory cycles; reduced LTV is associated with fetal compromise.
4) *Absolute Accelerations* — A straight tally tracking how many times the heart rate jumped by at least 15 beats for a minimum of 15 seconds.
5) *Total Decelerations* — An aggregate count of any notable fetal heart rate dips falling below the established resting baseline.
6) *Late Deceleration Flag* — Binary indicator of decelerations temporally lagging contraction peaks (key marker of uteroplacental insufficiency).
7) *Variable Deceleration Count* — Abrupt FHR drops, typically cord compression indicators.
8) *Approximate Entropy (ApEn)* — Non-linear regularity measure of FHR signal complexity.
9) *Sample Entropy (SampEn)* — Conditional probability of pattern continuation; complement to ApEn.
10) *UC Frequency* — Number of detectable contraction events per window.
11) *UC Amplitude* — Average peak-to-trough amplitude of contraction waveforms.
12) *FHR–UC Lag* — Cross-correlation lag at peak FHR-UC correlation, distinguishing early, variable, and late deceleration patterns.
13) *Valid Sample Density* — A ratio indicating the literal percentage of clean, non-gap segments available inside a given 20-minute inspection zone.

Missing values in tabular features (arising from recordings with insufficient signal quality for feature computation) were imputed using column-wise means computed from the training fold.

8) Step 8: Common Spatial Pattern (CSP) Feature Extraction: A complementary feature representation was derived using **Common Spatial Patterns (CSP)**, a technique originally developed for electroencephalographic (EEG) brain-computer interface (BCI) analysis, here applied as a novel modality for fetal monitoring. CSP identifies spatial filters that maximize the ratio of the variance of one class's signal to the other, applied to the two-channel (FHR, UC) input:

$$W = \mathrm{CSP}(X_{\mathrm{FHR}}, X_{\mathrm{UC}}) \in \mathbb{R}^d \tag{3}$$

where W comprises variance projections of FHR+UC interactions. This produces a **19-dimensional CSP feature vector** per window, providing a compact representation of the discriminative variance structure in the FHR–UC interaction that is not captured by either signal alone. CSP filters were fitted exclusively on the training fold within each cross-validation iteration to prevent data leakage.

C. Processed Dataset Summary

TABLE III
PROCESSED DATASET SPECIFICATIONS FOR NEUROFETAL AI TRAINING

Array File	Shape	Description
`X_fhr.npy`	$(N, 1200, 1)$	FHR windows at 1 Hz, 20 min
`X_uc.npy`	$(N, 1200, 1)$	UC windows at 1 Hz, 20 min
`X_tabular.npy`	$(N, 16)$	3 demographic + 13 signal-derived
`X_csp.npy`	$(N, 19)$	CSP variance features
`y.npy`	$(N,)$	Binary labels (0=Normal, 1=Comp.)
Approx. N (before augmentation)		$\sim$2,760
Pathological windows (before aug.)		$\sim$200 (7.25%)

D. TimeGAN Data Augmentation Strategy

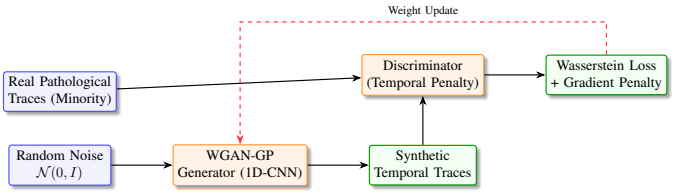

Fig. 1. TimeGAN data augmentation pipeline utilizing a Wasserstein GAN with Gradient Penalty (WGAN-GP) to synthesize physiologically plausible FHR and UC segments while preserving sequential dynamics.

The extreme class imbalance (7.25% pathological) necessitated a robust augmentation strategy. NeuroFetal AI v4.0

replaced the previously employed SMOTE with **TimeGAN**, a time-series generative adversarial network [15] implemented as a WGAN-GP architecture:

- **Generator Architecture**: 1D Transposed Convolution network trained exclusively on pathological FHR+UC paired windows.
- **Training Signal**: Wasserstein distance with gradient penalty (WGAN-GP, $\lambda = 10$) for training stability.
- **Output**: **1,410 synthetic pathological FHR+UC traces** ($3\times$ the original pathological count per fold).
- **Temporal Fidelity**: Unlike SMOTE, TimeGAN preserves physiologically meaningful sequential dynamics in generated traces— including late deceleration morphology and the temporal delay between contraction peaks and FHR nadir.
- **Application Scope**: Augmentation was applied independently per training fold inside the cross-validation loop, ensuring no synthetic samples from validation or test windows contaminated evaluation.

Following TimeGAN augmentation, standard time-series jitter, scaling, and time-warping transformations ($2\times$ expansion) were applied to the combined real+synthetic training set for additional regularization.

E. Cross-Validation Protocol

To ensure robust and unbiased evaluation, all model training and evaluation was performed under **Stratified 5-Fold Cross-Validation**:

- **Stratification**: Each fold preserves the $\approx$7.25% pathological class proportion.
- **Validation Ruleset**: We enforced strict Out-of-Fold (OOF) testing. In this setup, the algorithm generates exactly one blind prediction for every recorded window, guaranteeing that the reported metrics faithfully mirror unseen generalization performance.
- **Leakage prevention**: All data-dependent transformations (normalization statistics, CSP filter training, TimeGAN training, tabular imputation values) are computed exclusively from the training portion of each fold and applied to the held-out validation portion.
- **No independent held-out test set**: Given the limited dataset size (552 recordings), reserving a separate test partition would substantially reduce training data. The OOF paradigm provides an equivalent statistical guarantee without this sacrifice.
- **Deterministic Execution**: To permit perfect subsequent replication, we rigorously locked all pseudo-random generators using the initialization seed `42`.

IV. Proposed Architecture: AttentionFusionResNet

NeuroFetal AI (v5.0) introduces a tri-modal deep learning architecture, designated as **AttentionFusionResNet**. Most standard CTG classifiers lean entirely on the continuous FHR wave. In contrast, our setup weaves together three distinct layers of clinical reality: (1) Raw 1D Temporal FHR Signals, (2) Static Maternal Clinical Tabular Data, and (3) FHR-UC Spatial Pattern Features (CSP).

Architecturally, the model splits the data processing across three parallel feature extraction pipelines. These branches eventually merge at a *Cross-Modal Attention Fusion (CMAF)* hub before feeding into a final probabilistic classifier. We also integrated Monte Carlo (MC) Dropout specifically to quantify epistemic uncertainty.

The primary signal branch processes the 20-minute FHR sequence $\mathbf{X}_{\text{FHR}} \in \mathbb{R}^{1200\times 1}$. We built the temporal backbone entirely around a 1-Dimensional Residual Network (ResNet). We heavily adapted this backbone by injecting Squeeze-and-Excitation (SE) recalibration blocks [23] capped off with a Multi-Head Self-Attention routine.

The encoder begins with a large receptive field convolution (kernel size 7, stride 2) and max pooling to rapidly downsample the uninformative high-frequency noise. This is followed by three stages of cascading residual blocks. Let $\mathbf{x}_l$ denote the input to the l-th residual block. The block operation is defined as:

$$\mathbf{h}_l = \sigma(\text{BN}(\mathbf{W}_{l,2} * \sigma(\text{BN}(\mathbf{W}_{l,1} * \mathbf{x}_l)))) \tag{4}$$

$$\mathbf{x}_{l+1} = \text{ReLU}(\mathbf{x}_l + \text{SE}(\mathbf{h}_l)) \tag{5}$$

where $*$ denotes 1D convolution, BN is Batch Normalization, and $\text{SE}(\cdot)$ represents the channel-wise attention recalibration.

To prevent the model from memorizing the relatively small CTU-UHB dataset, we layered in **Stochastic Depth** (Spatial Dropout 1D) across the residual branches. This forces the network to dynamically drop entire communication channels during training with an increasing probability scale ($p \in [0.05, 0.15]$). Once the residual stages finish, a Temporal Multi-Head Attention block scans the data for long-range interactions—for instance, linking a subtle baseline shift at the 2-minute mark with a severe late deceleration hitting at minute 18. Finally, to collapse the extensive positional map into a tight descriptive bottleneck, a Global Average Pooling (GAP) layer squeezes the output into a single unified embedding vector $\mathbf{v}_{\text{FHR}} \in \mathbb{R}^{192}$.

A. Clinical Context and CSP Branches

Obstetric interpretation rarely happens in a vacuum. A doctor tracking a sudden deceleration handles the situation quite differently depending on whether the mother is 40 weeks along delivering her first child, versus 36 weeks along delivering her third.

The 16-dimensional tabular vector $\mathbf{X}_{\text{tab}}$ (comprising maternal demographics and extracted signal statistics such as Signal-to-Noise ratio and UC frequency) is processed through a two-layer Multi-Layer Perceptron (MLP) with ReLU activations and heavy dropout ($p = 0.3$) to yield the clinical embedding $\mathbf{v}_{\text{tab}} \in \mathbb{R}^{128}$.

Concurrently, the 19-dimensional Common Spatial Pattern vector $\mathbf{X}_{\text{CSP}}$, which encapsulates the discriminative variance between the FHR and UC contraction cycles, is processed through an identical, parallel MLP structure to yield $\mathbf{v}_{\text{CSP}} \in \mathbb{R}^{128}$.

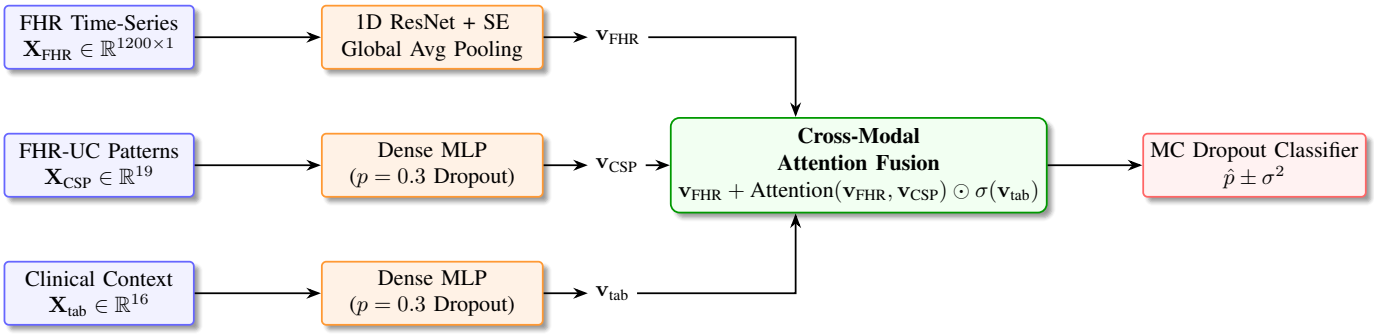

Fig. 2. Overall flow of the Tri-Modal AttentionFusionResNet Architecture. Raw FHR sequences are processed via 1D ResNet, while CSP variance and clinical metadata act as key/value patterns and gating contexts, respectively, converging at the Cross-Modal Attention Fusion (CMAF) layer.

B. Cross-Modal Attention Fusion (CMAF)

Existing late-fusion architectures often naively concatenate or multiply modal embeddings [25]. We propose a dynamic **Cross-Modal Attention Fusion (CMAF)** layer where the FHR temporal embedding attends to the FHR-UC spatial patterns, explicitly gated by the maternal clinical context.

Inside the CMAF unit, the data embeddings are dynamically projected out into Query ($\mathbf{Q}$), Key ($\mathbf{K}$), and Value ($\mathbf{V}$) coordinates:

$$\mathbf{Q} = \mathbf{W}_Q \mathbf{v}_{\text{FHR}} \tag{6}$$

$$\mathbf{K} = \mathbf{W}_K \mathbf{v}_{\text{CSP}} \tag{7}$$

$$\mathbf{V} = \mathbf{W}_V \mathbf{v}_{\text{CSP}} \tag{8}$$

Cross-modal attention scores are computed via scaled dot-product:

$$\mathbf{A} = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \tag{9}$$

The attention output is gated by a learned, sigmoid-activated projection of the clinical tabular data:

$$\mathbf{G}_{\text{clinical}} = \sigma(\mathbf{W}_G \mathbf{v}_{\text{tab}}) \tag{10}$$

$$\mathbf{v}_{\text{fusion}} = \text{LayerNorm}(\mathbf{v}_{\text{FHR}} + \text{Dropout}(\mathbf{A} \odot \mathbf{G}_{\text{clinical}})) \tag{11}$$

This essentially creates a gating mechanism. It permits the main network to on-the-fly adjust exactly how much importance it places on the spatial contraction patterns, strictly dictated by the mother's unique clinical risk profile.

C. Classification and Uncertainty Head

The resulting fused embedding $\mathbf{v}_{\text{fusion}}$ acts as the input for a classification block built from two stacked dense layers. To capture robust diagnostic doubt and model uncertainty logic, the architecture implements the highly reliable Monte Carlo (MC) Dropout bayesian approximation technique [17].

While standard dropout is normally shut off during testing, we deliberately leave the target dropout layers ($p = 0.3$) open during active inference. When evaluating a new patient trace in a clinical setting, the system runs $T = 20$ randomized forward passes. The architecture's final predictive probability $\hat{p}$ simply averages those passes together, while tracking the predictive variance σ^2 to measure how much diagnostic doubt exists. This variance is crucial for instantly flagging borderline cases that demand human intervention.

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